

Extracting Features from EEG Signals

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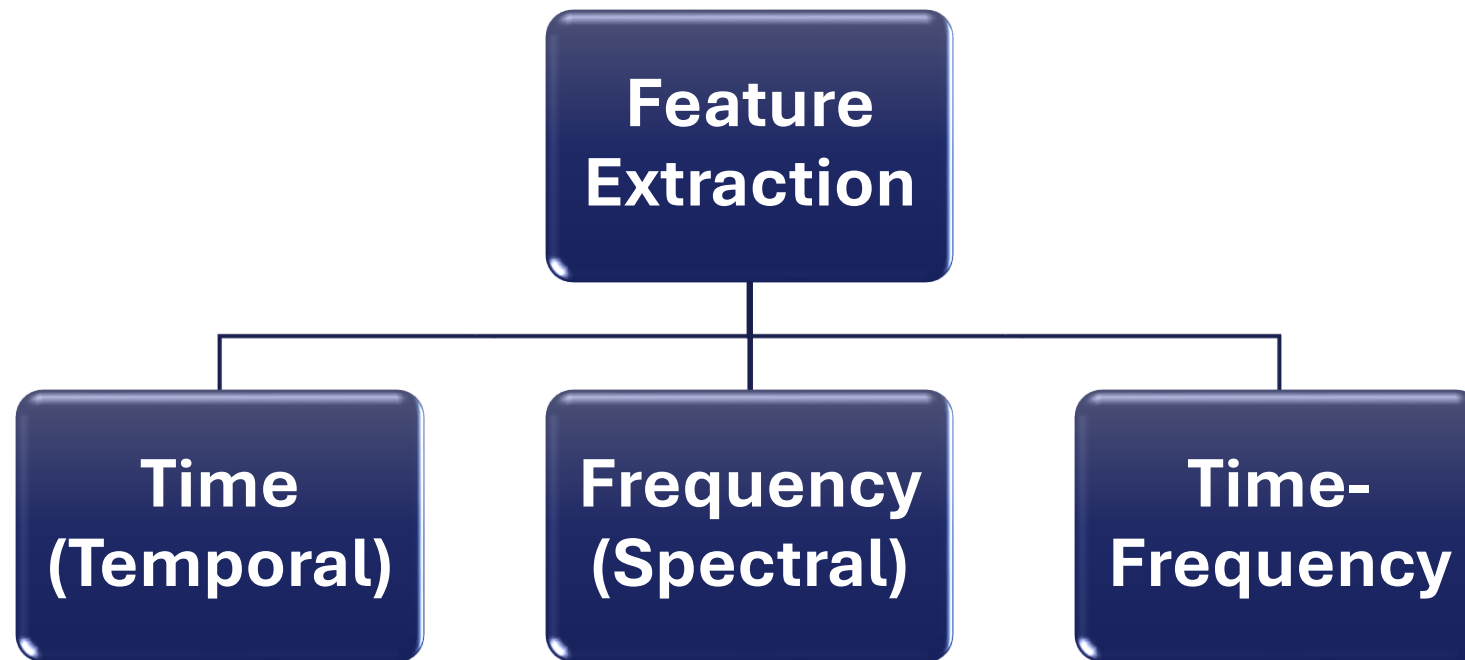
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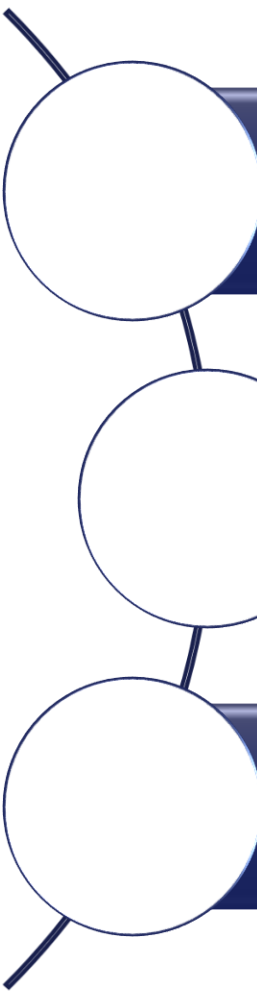
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Computational intelligence for EEG-based BCI

The BCI uses computational intelligence to extract essential features from the pre-processed EEG signals and feed the feature vector into a classifier to control the BCI output.



Features extraction



The BCI features extraction step transforms the raw EEG data into a feature vector for optimal classification.

If the extracted features do not describe the neuro-physiological signal, inaccurate classification may result.

To optimize the BCI's classification accuracy, a proper feature vector must be extracted from the raw EEG signal.

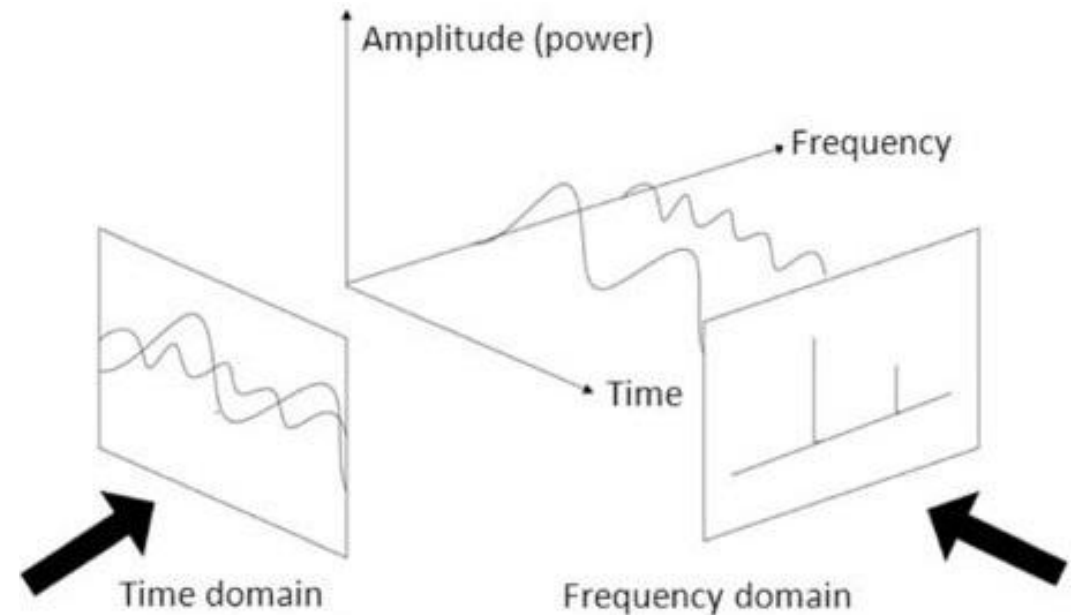
Signal representation in time versus frequency domain

Time domain

- **Representation:** A signal is visualized as a function of time. The x-axis represents time, and the y-axis represents the signal's value at that time.
- **What it represents:** Shows how the signal's amplitude (or value) changes over time.
- **Time to frequency:** *Fourier Transform (FT)*. This mathematical operation decomposes a signal into its frequency components.

Frequency domain

- **Representation:** A signal is visualized as a function of frequency. The x-axis represents frequency, and the y-axis represents the amplitude of each frequency component present in the signal.
- **What it represents:** Shows the composition of the signal in terms of its constituent frequencies.
- **Frequency to time:** *Inverse Fourier Transform (IFT)*. This mathematical operation reconstructs a signal from its frequency components.



Fourier transform versus power spectral density in EEG

Feature	Fourier Transform (FT)	Power Spectral Density (PSD)
Purpose	Frequency decomposition	Power distribution
Stationarity Assumption	Assumes stationarity (such as a sine wave)	Can handle non-stationary signals (such as task-related changes in EEG)
Usage	Identifying dominant frequencies	Power estimation (e.g., studying brain activity)
Limitations	Not suitable for non-stationary signals	Will require windowing for non-stationary signals

What about the time-frequency representation (TFR)?

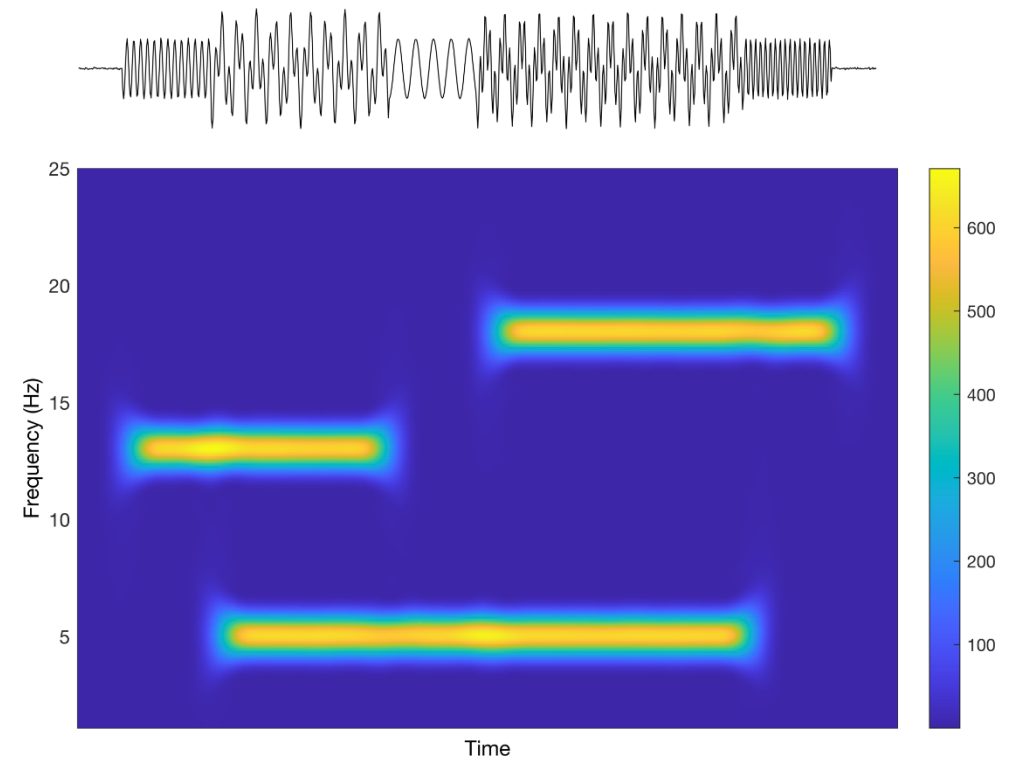
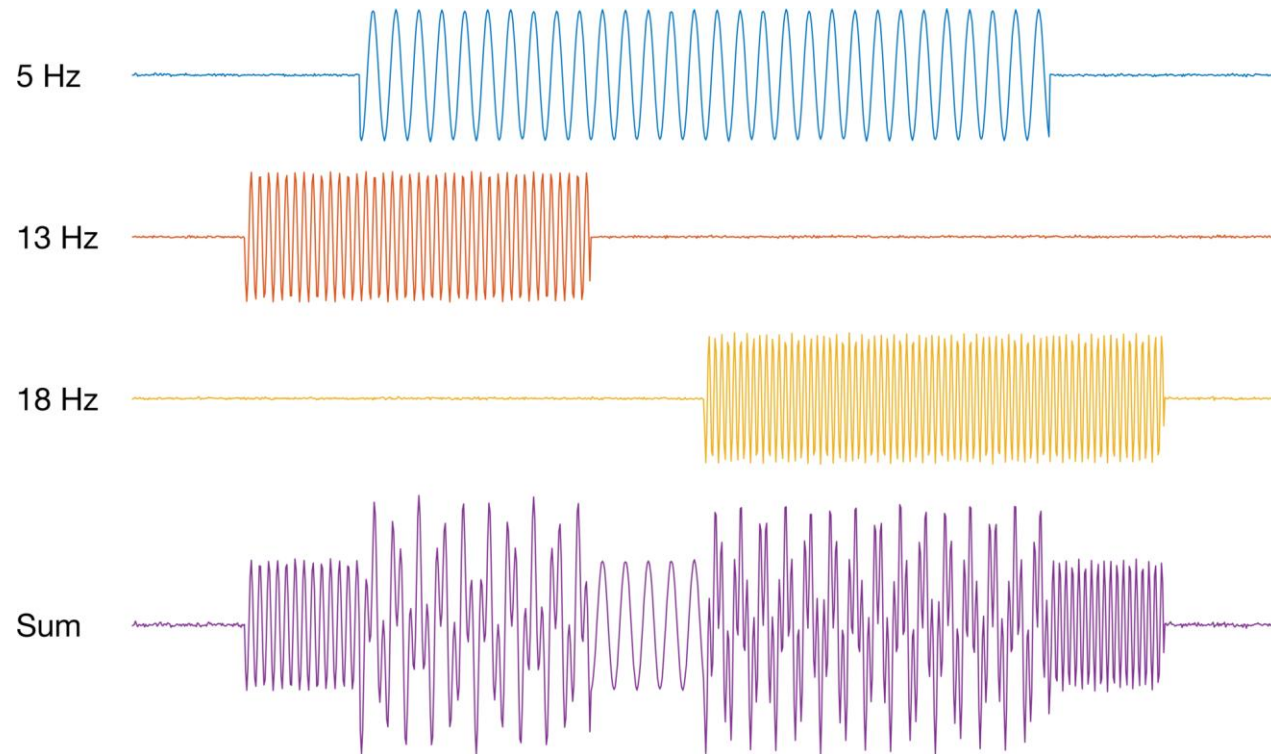
A bridge between time and frequency domains

- TFR is a method that combines the **advantages of both the time and frequency** domains for analyzing signals.
- It provides a visual **representation of how the frequency content of a signal changes over time.**

Benefits of time-frequency analysis:

- **Analysis of non-stationary signals:** Many real-world signals, such as speech, music, and biomedical signals, are **non-stationary**, meaning their frequency content changes over time.
- **Signal classification:** TFRs can be used to classify signals based on their time-frequency characteristics.

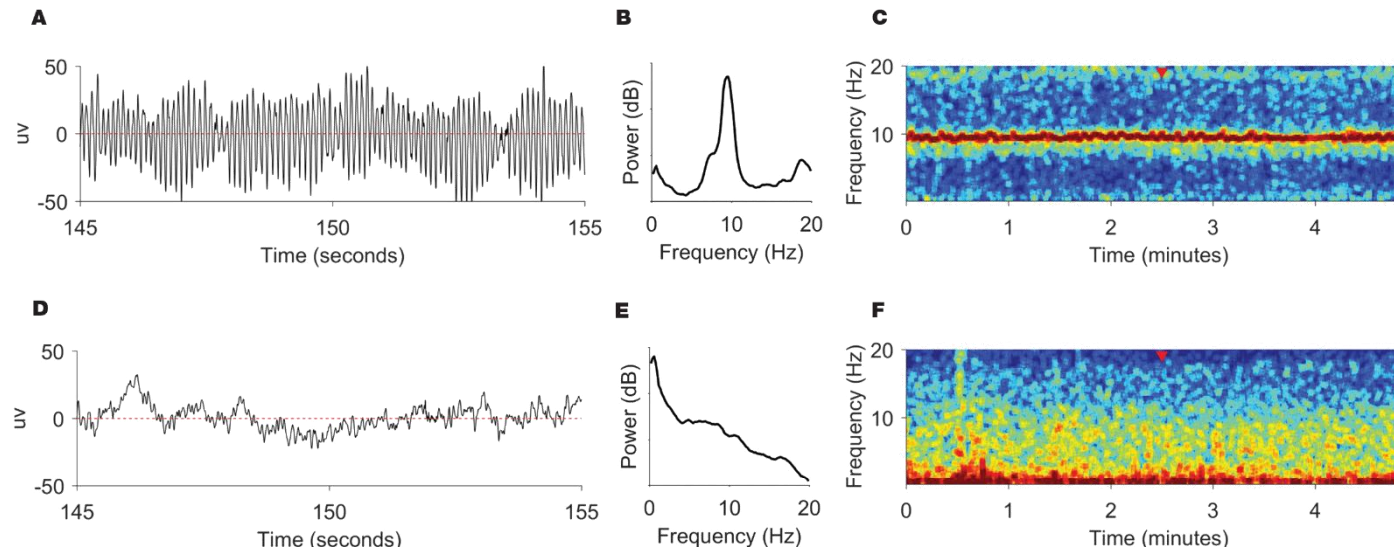
TFR example



Inspection of the TFR results - Spectrogram

It is a common TFR that uses a sliding window to analyze the frequency content of a signal at different time intervals.

The **x-axis** represents **time**, the **y-axis** represents **frequency**, and the **color of intensity** represents the **amplitude of the frequency components**.



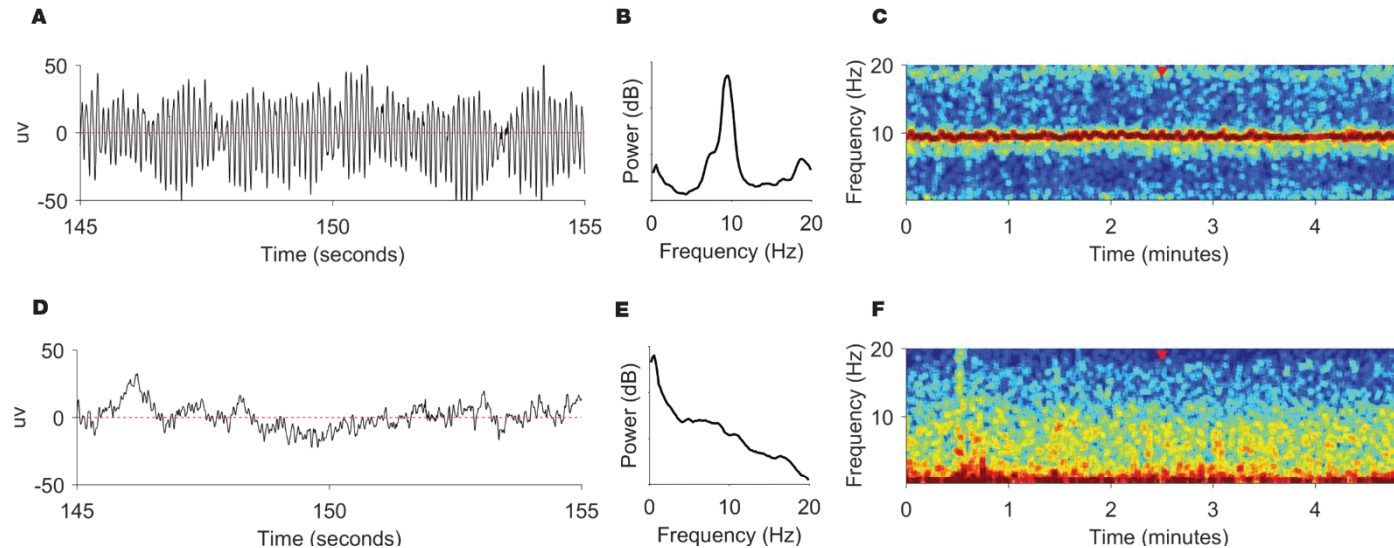
Localized

Distributed

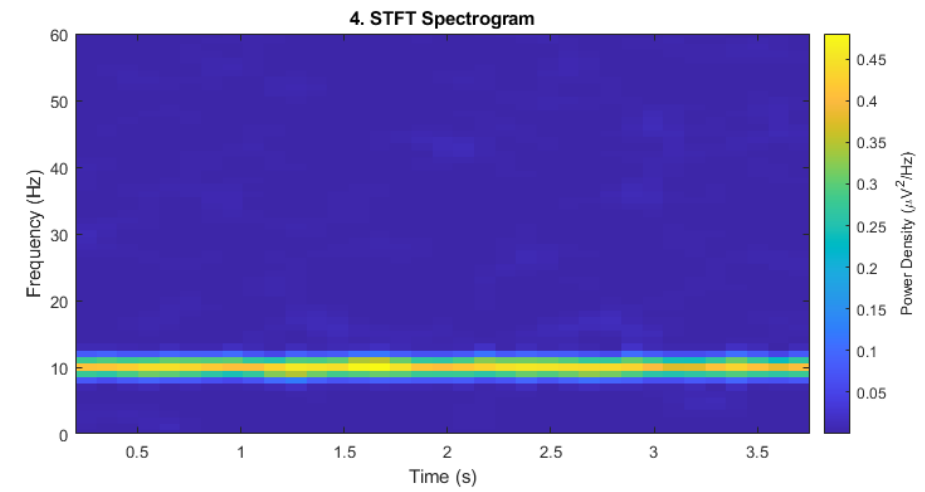
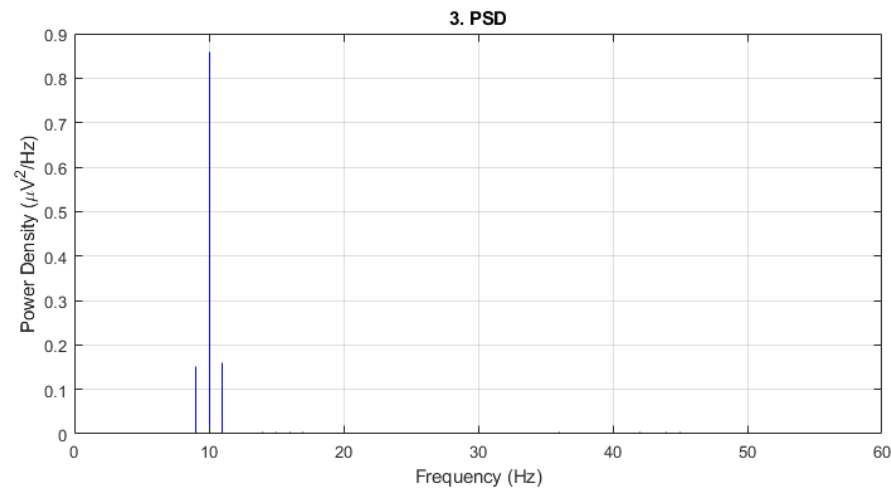
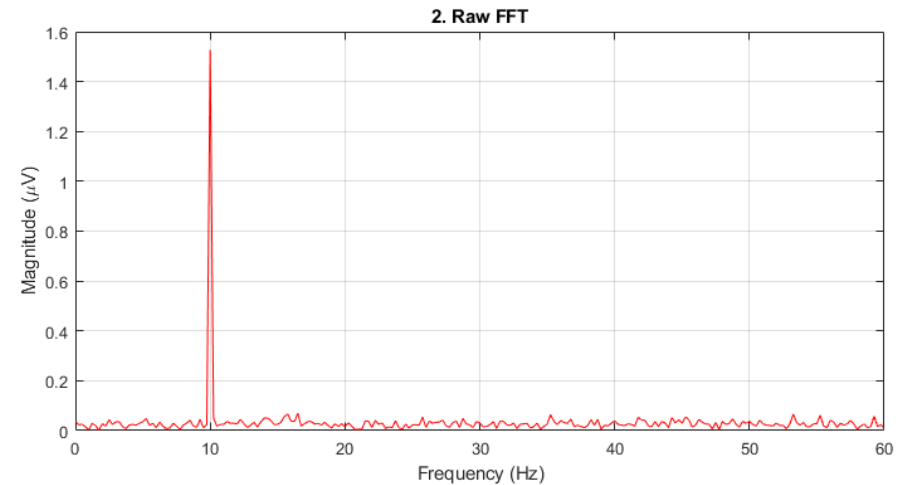
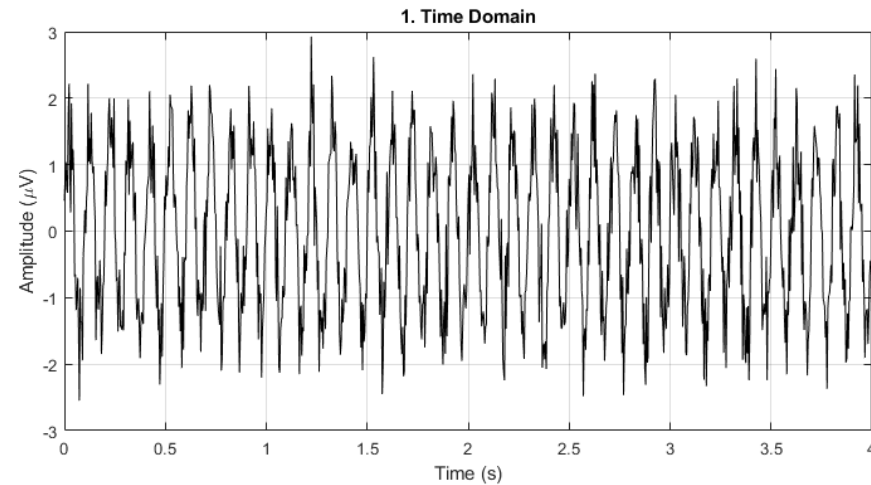
Inspection of the TFR results - Windowing

The choice of window function affects the resolution of the TFR:

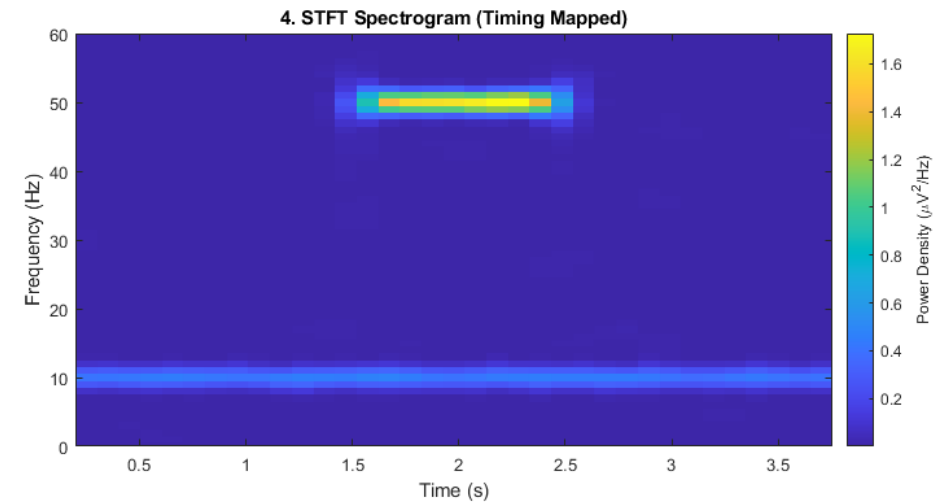
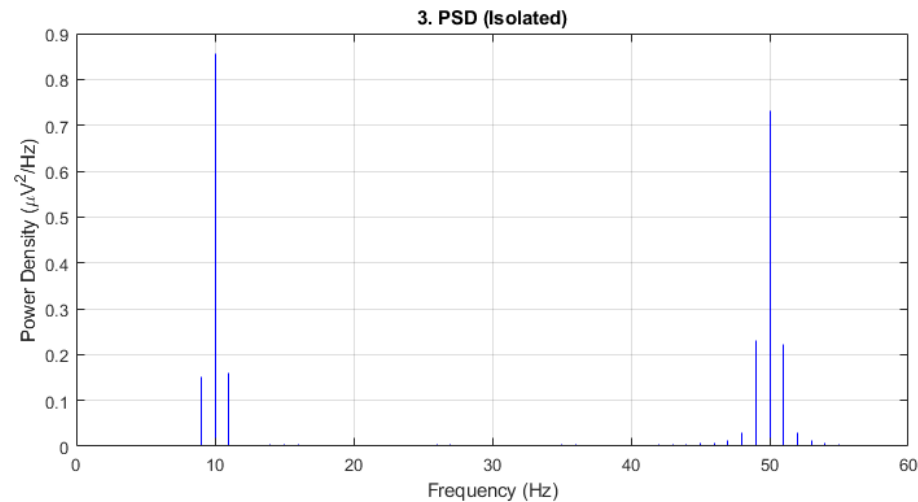
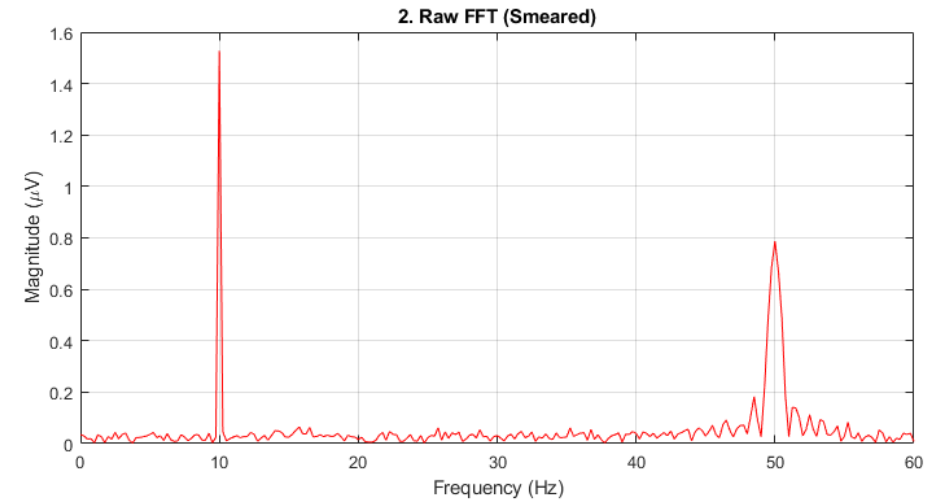
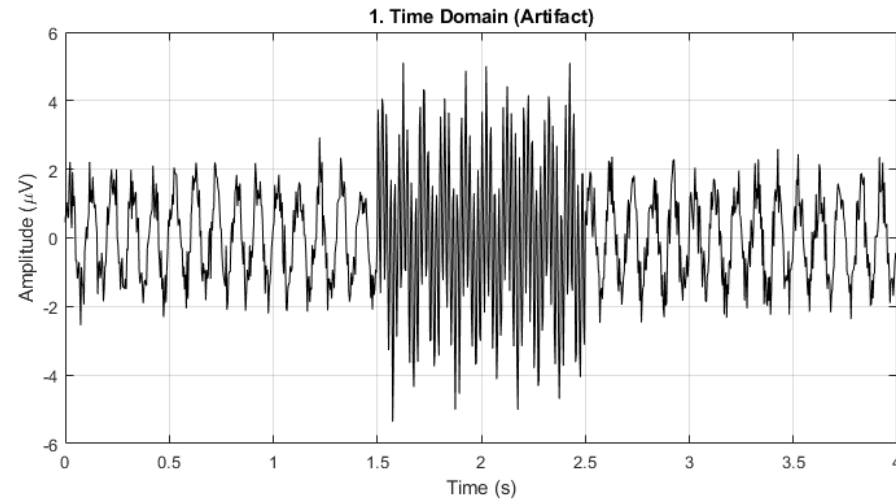
- A **wider window** provides better frequency resolution but poorer time resolution (slower system).
- A **narrower window** provides better time resolution (faster system) but poorer frequency resolution.



PSD or FFT for feature extraction – Stationary signal



PSD or FFT for feature extraction – Nonstationary signal



Windowed PSD vs. TFR analysis

Windowed spectral features (PSD/1D arrays)

Dimensionality: Compresses a segment of time into a simple, flat **1D array of power values** (e.g., [Alpha_Power, Beta_Power]).

Computational load: Extremely lightweight and fast.

Machine learning fit: Ideal for traditional, **fast classifiers** (like LDA or SVM) that can be trained and executed quickly on standard hardware.

Best for: Rapid, real-time tracking of general states such as mental fatigue or basic motor imagery.

Time-frequency analysis (STFT/2D matrices)

Dimensionality: Generates a massive, **dense 2D matrix (spectrogram)** mapping exact frequencies to split-second timing.

Computational load: Highly resource-intensive.

Machine learning fit: Requires complex **deep learning models** (like CNNs) designed to process the data array as a visual "image."

Best for: Spatially map and pinpoint the exact millisecond a dynamic brainwave shifts or a temporary artifact occurs.

Time-locked versus phase-locked activity

Time-locked and phase-locked

- **Yes**, in evoked activities.

Time-locked and not-phase-locked

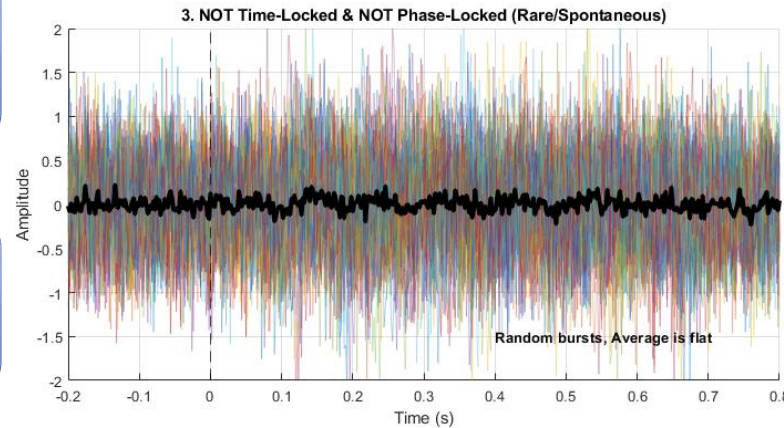
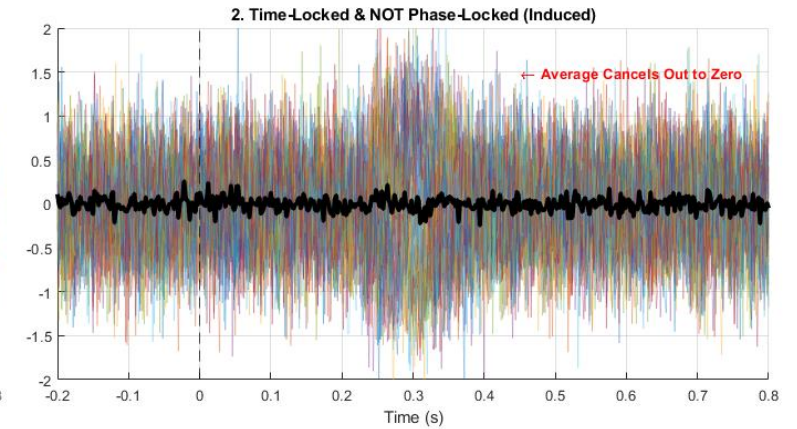
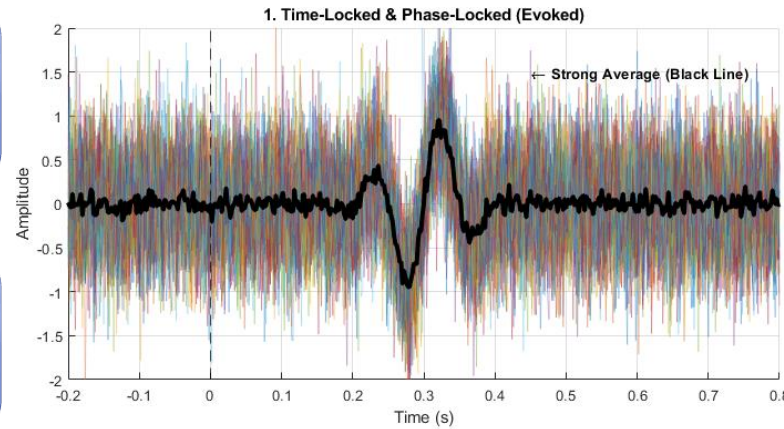
- **Yes**, in induced activities.

Not-time-locked and not-phase-locked

- **Yes**, but very rare and complex cases.

Not-time-locked and phase-locked

- **No**, not mathematically possible.



4. NOT Time-Locked & Phase-Locked

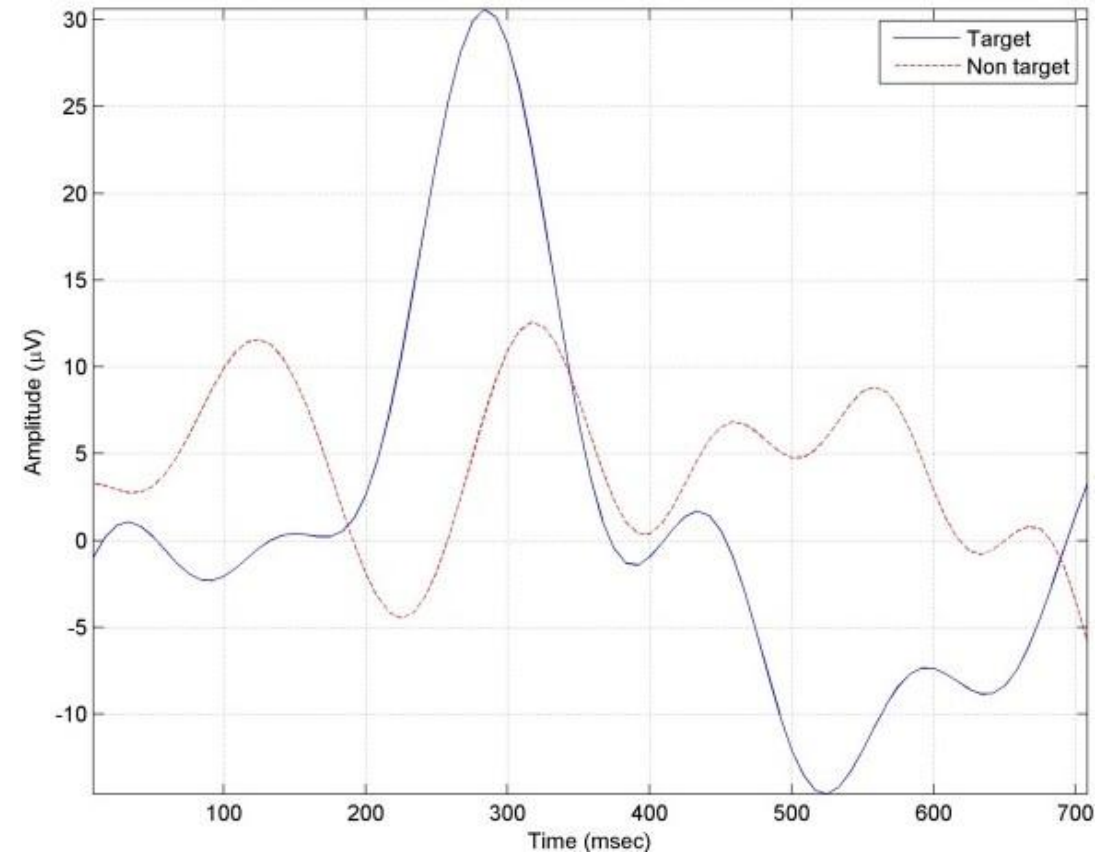
Not Mathematically Possible

(If the wave shifts in time relative to the stimulus, its phase relative to the stimulus onset inherently changes)

Time (temporal) features

This method describes the EEG-based BCI as a neurophysiological signal with a precise and specific time signature.

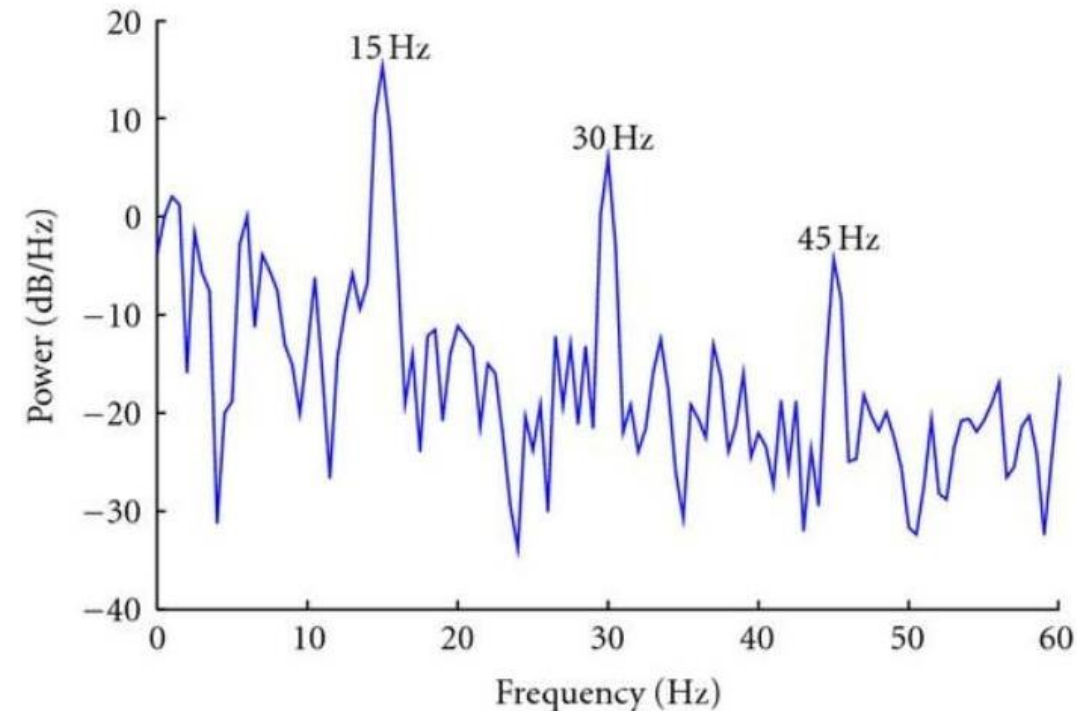
A feature based on a signal amplitude is the simplest temporal information, **P300** is an example of a temporal feature.



Frequency (spectral) features

One of the most basic methods for tracking the amplitude modulation at a particular frequency is using the band-power feature extraction method.

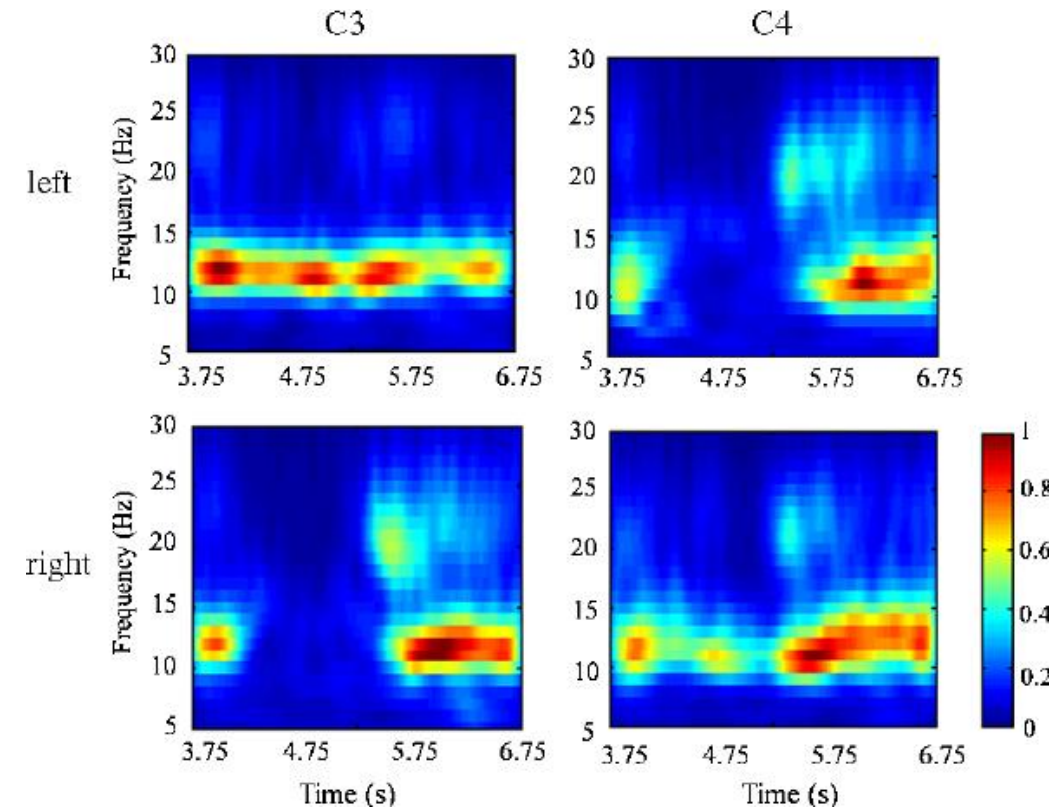
It is practical to isolate the frequency of interest by filtering the signal with a band-pass filter, **SSVEP** is an example of a frequency feature.



Time-frequency features

Instead of using features from time or frequency domain alone, the combination of time-frequency features extractor method has also been used to design BCI.

Wavelet analysis is an example of a time-frequency feature extraction method for **MI** signals.



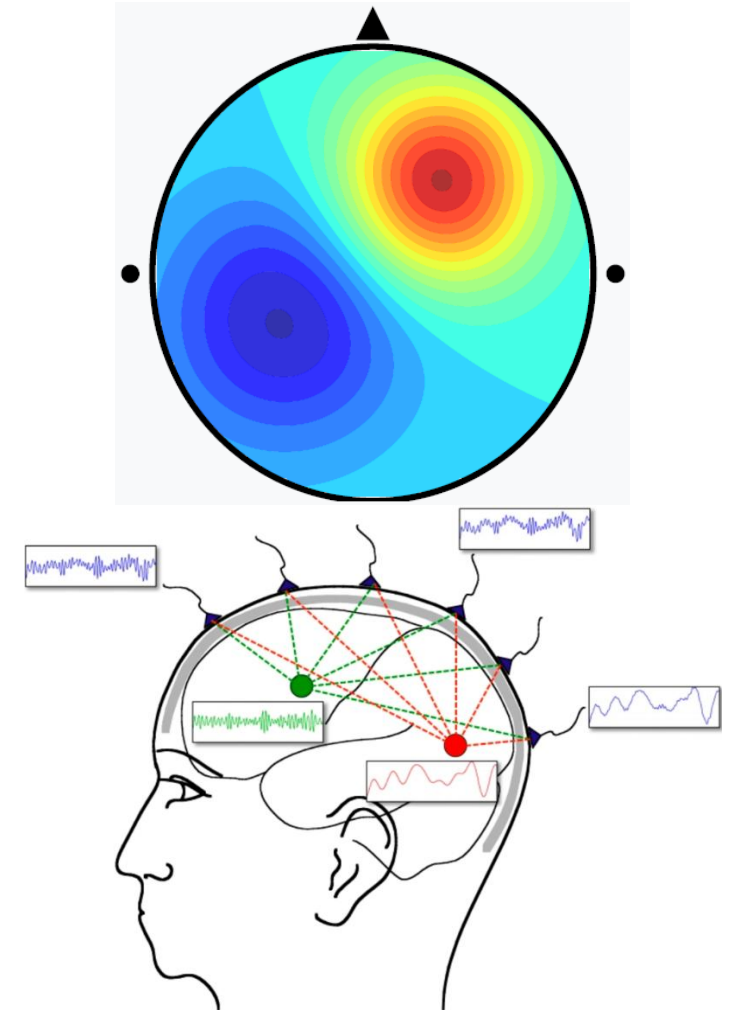
Spatial features in EEG

What are they?

- Spatial features capture the **physical distribution** and relationship of electrical activity across different scalp electrodes.

Why do we need them?

- Electrical signals smear and spread as they travel through the cerebrospinal fluid, skull, and scalp.
- A single electrode captures a blurred blend of multiple underlying brain sources and artifacts.
- Spatial features help map signals back to their true neuroanatomical origins, improving classifier accuracy.
- Typical use is for differentiating the SMR's motor sources.



Evoked versus induced activity – Origin

Evoked response

It is believed to be the result of coherent firing of neurons in response to the stimulus.

Evoked response paradigms are suited for studying **bottom-up** (react to external) sensory processes.

Induced response

It is thought to be related to higher order processes of the brain which are not directly related to the stimulus but are caused indirectly by nonlinear interactions of neurons following the stimulus.

Induced responses reflect **top-down** (initiate from internal) higher order brain functions.

Evoked versus induced activity – Behaviour

Evoked activity

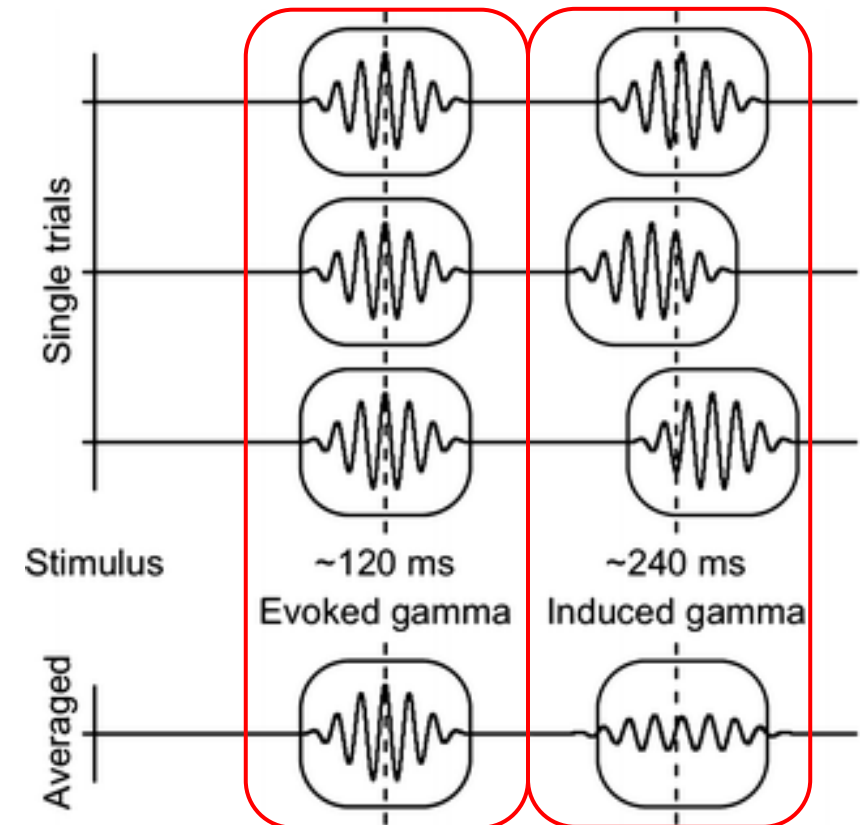
The evoked response is the **brain response to the presented stimulus** such that peaks occur at fixed times (latencies) following the stimuli.

Time-domain averaging can be used to enhance these responses and suppress the usually overwhelming background EEG activity which hides the response in the raw signals.

Induced activity

The induced response is the **brain response to the presented stimulus** such that there may be an increase or decrease in the amplitude of the signal at a specific time following a stimulus.

This increase or decrease in amplitude does not consistently involve the same phase angle in all stimulus presentations and that is why these amplitudes may be **cancelled out using time-domain averaging**.



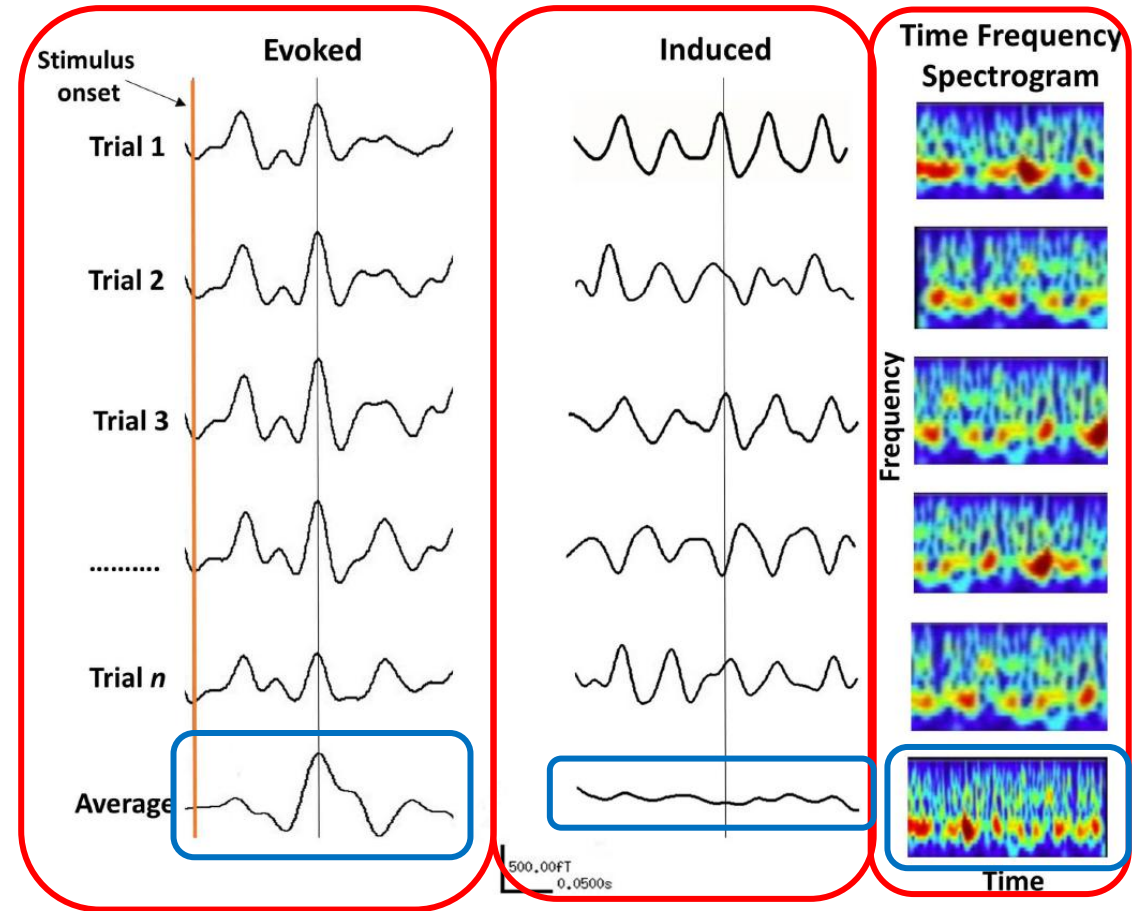
Evoked versus induced activity - Processing

The **evoked activity** (left) is the linear time average of many short trials whose responses are **time and phase-locked** to the onset of a stimulus.

For **induced responses** (middle), linear time averaging will remove the effect of interest due to trials being **not phase-locked while being time-locked to the stimulus**.

The information within induced responses is obtained by evaluating the frequency spectrum of each trial (right) overtime, which is then averaged.

The frequency information of individual trials is thus retained (bottom right panel).



Relation between evoked activity and BCIs

Neural activity that is strictly **time-locked** and **phase-locked** to an external sensory stimulus (visual, auditory, or tactile).

- Because responses align precisely in time across repetitions, **trials are averaged in the time domain** to cancel background noise and expose distinct voltage peaks.

Primary BCI paradigms

- **Transient ERPs**
 - **P300**: Positive peak at ~300 ms triggered by rare target recognition (e.g., P300 visual spellers).
 - **ERN & N-family (N100, N200, N400)**: Neural markers for error detection, semantic mismatch, and cognitive processing.
- **Steady-state evoked potentials**
 - **SSVEP / ASSR**: Continuous frequency entrainment where the brain's sensory cortex physically locks its firing rate to an external flicker (e.g., 15 Hz flashing target).

Relation between induced activity and BCIs

Neural activity that reflects self-generated mental states and is not phase-locked to precise external triggers.

- Because phase varies across trials, standard time-domain averaging cancels out the signal. Instead, **time-frequency analysis (EROs)** and **PSD** are used to track continuous increases or decreases in frequency band power.

Primary BCI paradigms

- **EROs**
 - **SMR**: Imagining physical movement triggers ERD (power drop in mu and beta bands) followed by ERS (power rebound), driving robotic prosthetics and cursor navigation.
 - **Cognitive and mental imagery**: Tracking frontal beta and parietal alpha shifts for passive BCIs, mental workload monitoring, and neuro-adaptive systems.

Feature transformation

What it is

The mathematical projection of the feature matrix into an entirely new **spatial representation**.

Why doing it

Unscrambles overlapping signal sources and maximizes class separability when standard extracted features are too deeply entangled or corrupted by artifacts.

How it works

Utilizes spatial algorithms to recombine the data, such as Principal Component Analysis (PCA) for **variance-based dimensionality reduction**, Independent Component Analysis (ICA) for **blind source separation**, or Common Spatial Patterns (CSP) to **maximize the variance between two distinct cognitive states**.

Feature selection

What it is

The **strategic filtering** of your extracted dataset to keep only the most robust, relevant variables.

Why doing it

Prevents the machine learning model from memorizing noise (overfitting), eliminates redundant data, and significantly reduces the computational load of the pipeline.

How it works

Employs **statistical evaluators** to mathematically score every feature's ability to separate target classes, discarding the low-performing metrics.

Integrating the feature engineering pipeline

Step 1: Extract:

- First, translate the raw physiological time-series data into a quantifiable mathematical matrix.

Step 2: Transform:

- Next, project those newly extracted variables into a different spatial domain to untangle overlapping signal sources and maximize the mathematical boundaries between classes. ***Only works with higher number of electrodes though.***

Step 3: Select:

- Finally, mathematically prune the dataset, discarding redundant data and keeping only the absolute highest-performing raw or transformed features.

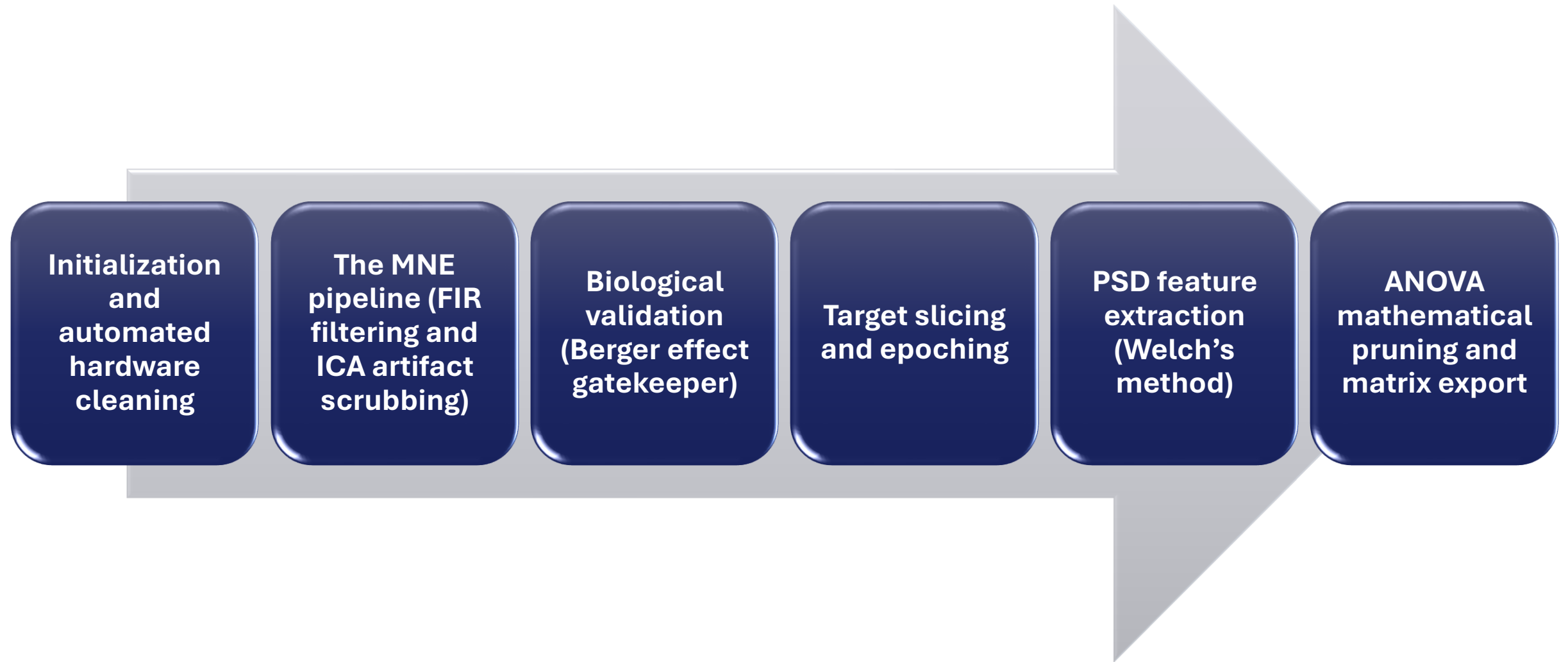
Muse 2 offline pipeline

Feature extraction

Required libraries and dependencies

<code>os, sys, and glob</code>	Automates the traversal of subject directories and the batch discovery of raw CSV files.
<code>pandas and numpy</code>	Handles the structuring of the global feature matrix and performs high-speed numerical array calculations for signal integration.
<code>scipy.signal</code>	Executes the mathematical transformations, including filtering and PSD calculation.
<code>sklearn.feature_selection</code>	Supplies the functions to perform statistical ANOVA F-tests for mathematical feature pruning.
<code>tkinter</code>	Generates the graphical login portal to securely route the operator to the correct existing subject profile.
<code>mne</code>	The industry-standard library for neurophysiological data analysis. We utilize it for advanced signal reconstruction, spatial unmixing, and mathematical artifact scrubbing.

Pipeline overview



Initialization and automated hardware cleaning

Load raw 4-channel EEG data (TP9, AF7, AF8, TP10) from the subject's CSV files.



Scan the matrix for hardware-triggered safety flags (*Artifact_Flag* and *Rhythm_Flag*).



Automatically drop any corrupted data points flagged by the headset during the live recording.

DSP filtering and ICA artifact scrubbing

Convert the continuous raw data into MNE-Python RawArray objects.



Apply FIR filters: A 50Hz Notch filter to remove electrical line noise, and a 1-40Hz Bandpass filter.



Run ICA to unmix the spatial signals, using the AF7 channel as a proxy EOG.



Mathematically zero out ocular artifacts (blinks) and reconstruct a clean, continuous brainwave signal.

Practical implementation

Zero-phase filtering

The script uses **FIR (Finite Impulse Response)** filtering. This runs the filter both forwards and backwards, ensuring zero phase distortion (time-shifting) of the brainwaves.

The bandpass window

A 1 to 40 Hz window is strictly applied. This successfully eliminates DC drift (sweat artifacts below 1 Hz) and filters out high-frequency electromyography (EMG muscle noise above 40 Hz).

Epoch length

Data is segmented into exactly 2.0-second blocks, providing a sufficient time window to accurately capture the slower Alpha rhythms.

Repair vs. discard

Instead of simply dropping heavily contaminated files, we will **mathematically repair** them. By separating the raw signal into independent spatial components, we can surgically delete biological noise while preserving the underlying brainwaves.

Biological validation gatekeeper

Isolate Marker 1 and Marker 2 (baseline data) from the newly scrubbed continuous signal.



Calculate the Berger effect across posterior channels (TP9, TP10) using Welch's method.



Require a **>1.2x Alpha power surge** to pass the gatekeeper. Files failing this physiological threshold are dropped due to poor cortical contact.

Target slicing and epoching

Slice the validated, continuous dataset to retain only the target ML tasks (Marker 3 and Marker 4).



Segment the continuous target data into discrete, uniform 2.0-second epochs based on the 256Hz sampling rate.



Assign binary classification labels (0 for Relax, 1 for Focus) to each extracted epoch.

The theory of cognitive states

Alpha waves (8–12 Hz)

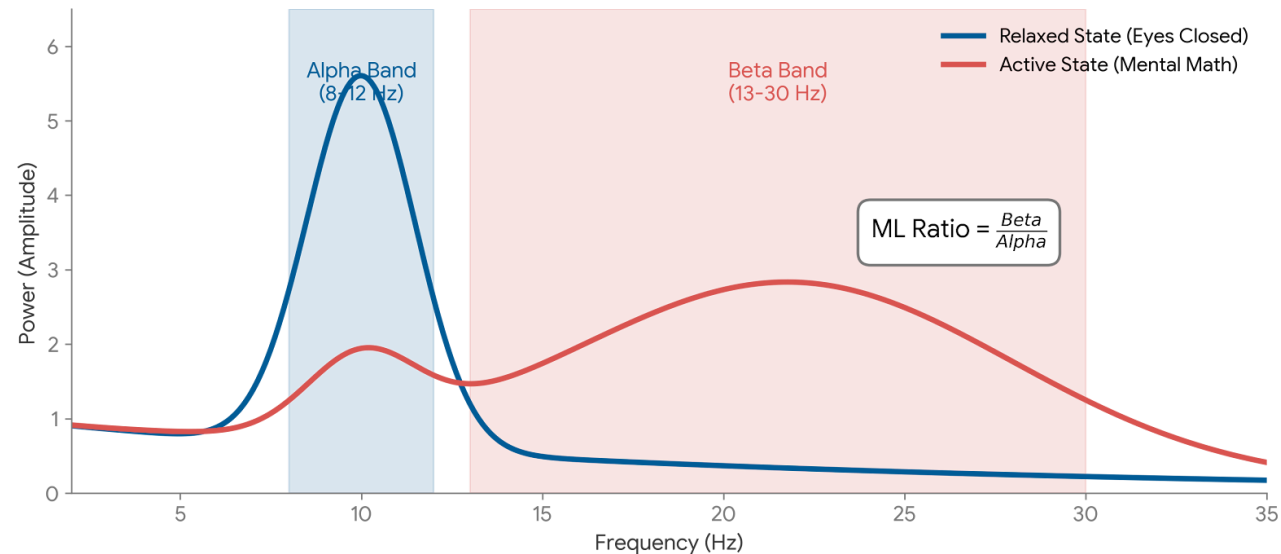
The brain's idling state. Surges in the **temporal lobes** when the eyes are closed and the mind is relaxed.

Beta waves (13–30 Hz)

The brain's processing state. Dominates the **prefrontal cortex** during active reading, stress, or mental math.

The ML target

Training classifiers not on raw voltage spikes, but on the continuous **ratio of Beta power divided by Alpha power**.



Feature extraction strategy

What was extracted?

- The script transforms raw time-series data into the frequency domain using **Welch's method to calculate PSD**.
- It then isolates the Alpha band (8-12 Hz) and Beta band (13-30 Hz) across all four Muse channels (TP9, AF7, AF8, TP10).
- Instead of just taking the absolute integral power, the **absolute power is divided by the band's width to output a normalized "true PSD density per Hz"**.

Why these specific features?

- Alpha and Beta are the primary established biomarkers for the experimental protocol. Alpha suppression defines the resting baseline (Marker 3), while sustained Beta power defines the active cognitive load (Marker 4).
- Beta is a wide band (17 Hz span) while Alpha is a narrow band (4 Hz span). Normalizing the power to a "per Hz" density ensures that wider bands do not artificially overpower narrower bands when fed into the machine learning classifier.

PSD feature extraction

Process each 2.0-second epoch independently.



Utilize Welch's method to calculate the PSD for each individual channel.



Extract the absolute power for the Alpha (8-13 Hz) and Beta (13-30 Hz) frequency bands, then normalize power per Hz.



Calculate the Beta/Alpha ratio for each electrode. In neurophysiology, the Beta/Alpha ratio is often referred to as an "**engagement index**."



By turning absolute power into a ratio, you mathematically normalize baseline differences between subjects and create a single metric that directly tracks cognitive workload.

Selection and matrix export

Compile all extracted PSD features into a single, fused dataset.



Execute ANOVA (Analysis of Variance) to statistically score the separability of the features.



Select the top 2 mathematical features possessing the highest F-Scores.



Export the finalized, pruned feature matrix to a CSV for offline classifier training.

Quality and feature selection report example

HERBioLab - Biological Validation Quality Report

Subject ID: 001 | Task: MATH

Note: Frontal artifacts are now scrubbed mathematically via MNE ICA.

=====

[muse_raw_math_run1.csv]  Passed: Alpha surged by 1.85x.

[muse_raw_math_run2.csv]  Passed: Alpha surged by 1.32x.

[muse_raw_math_run3.csv]  Failed: Alpha ratio was only 1.05x (Poor cortical contact).

=====

PASS 2: ANOVA FEATURE SELECTION AUDIT

=====

Top 2 Features Selected (Higher F-Score = Better Class Separability):

1. TP9_Beta_Alpha_Ratio (F-Score: 45.20)

2. TP10_Beta_Alpha_Ratio (F-Score: 38.75)

How to read the quality and feature selection report

The Alpha surge (e.g., 1.85x): The script compares the Alpha band power in the rear electrodes (TP9 and TP10) between the baseline resting marker (Marker 1) and the active marker (Marker 2). A natural biological brain will show an Alpha surge when the user closes their eyes or relaxes.

The 1.2x threshold: The script enforces a strict rule: if the Alpha power does not surge by at least 1.2x (20% increase), the file fails the check. In the example above, run3.csv failed at 1.05x, meaning the headset was likely loose or the participant wasn't actually relaxing, and the script will automatically exclude it from machine learning.

Top 2 features selected: Out of the 4 available electrodes, the algorithm selects exactly the top 2 features to prevent the machine learning model from getting confused by noisy channels.

F-Score (e.g., 45.20): This is a statistical score representing “class separability”. A higher F-Score means that the mathematical difference between the resting phase (Marker 3) and active task (Marker 4) is highly distinct on that specific electrode. In this example, the left rear electrode (TP9_Beta_Alpha_Ratio) provided the clearest, most usable signal for the classifier.

*Thank
You!*



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